

Shrey Patel

US Citizen | 414-539-9402 | shreyp2305@gmail.com | linkedin.com/in/shreyp2305 | github.com/shreyp2305

Education

Bachelor of Science, Computer Science

May 2025

Virginia Tech

GPA: 3.86

Relevant Courses: Data Structures and Algorithms, Operating Systems, Computer Architecture, Database Management Systems, Cloud Software Development, Intro to AI, Machine Learning, Urban Visualization

Technical Skills

Languages: Java, Python, TypeScript/JavaScript, C/C++, C#, Kotlin, Go, SQL, HTML, CSS

Frameworks & Libraries: Spring, .NET, Flask, React, Node.js, LangChain, JUnit, Cypress, Jest

Messaging & Databases: RabbitMQ, Kafka, PostgreSQL, MongoDB, Redis

Cloud/DevOps: AWS, Docker, Kubernetes, Jenkins, Terraform, Git

Experience

Full-Stack Software Engineer

Oct 2025 – Present

Integration Innovation

Huntsville, AL

- Built a modular plugin in Typescript to integrate autonomous mission coordination services into a command-and-control platform, enabling real-time vehicle management across 13 platforms.
- Enabled platform ownership transfers across distributed clients by developing a thread-safe coordination service that supported 50+ concurrent ownership negotiations using atomic state transitions and automated reconciliation.
- Stabilized geospatial visualization system from feature and layer corruption, by architecting a serialized pipeline to coordinate asynchronous layer and feature updates.
- Reduced manual QA effort 90% by building an end-to-end testing framework with Cypress and Cucumber, automating 23 mission-critical user flows and directly mapping tests to operational acceptance criteria.

Full-Stack Software Engineer Intern

Aug 2024 – Feb 2025

Peraton

Blacksburg, VA

- Built event-driven Java microservices to ingest real-time satellite telemetry across 5 ground stations, enabling 100+ concurrent operator sessions.
- Migrated 20+ TypeScript/React components to a shared styled-component library, eliminating redundant CSS and standardizing the UI across 4 screens.
- Implemented subsystem-specific RabbitMQ channels as part of a system re-architecture, cutting message queuing latency by ~70% (800 ms → 240 ms).

Full-Stack Software Engineer Intern

May 2024 – Aug 2024

CACI

Reston, VA

- Reduced manual reporting time from 3 days to under 15 minutes by developing dynamic UI dashboards that gave analysts self-service access to financial and user activity data, eliminating dependency on monthly data requests.
- Revamped a data validation service by optimizing PL/SQL procedures and adding composite indexes, cutting average query latency from 1 seconds to under 200ms across tables with 8+ million rows.

Full-Stack Software Engineer Intern

May 2023 – Aug 2023

Ventura

Reston, VA

- Scaled Node.js APIs serving 1,000+ daily users by transitioning high-traffic endpoints to a Redis cache-aside architecture, cutting Time-to-First-Byte (TTFB) by 90% and increasing throughput 3x.
- Raised code coverage from 67% to 97% by authoring 30+ Jest unit tests mocking API calls and React component interactions.
- Identified over-provisioned Kubernetes workloads by developing a performance testing suite integrated with Jenkins, reducing CPU usage by 25% and memory/pod counts by 15%.

Projects

Personal Portfolio Website [🔗](#) | NextJS, OpenAI, LangChain, Vercel AI SDK, Redis

Aug 2025 – Present

- Implemented a RAG pipeline using LangChain, embedding personal context into a vector store and routing semantic queries through GPT-5 via Vercel's AI SDK.
- Implemented cache-aware querying with a serverless Redis instance to reduce latency and operational costs.

Leadership/Extracurricular

FIRST Robotics Mentor | 20 Students

Aug 2022 – Present

Undergraduate Teaching Assistant | CS4604 Database Management Systems

Jan 2024 – May 2024

Resident Advisor | 150 Residents

Jan 2022 – May 2023